

Delays in the construction of power plants from electricity auctions in Brazil

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ABSTRACT

Brazil has a long-running experience with auctions as an instrument to procure electricity both from renewable and non-renewable sources. We build a database containing project-level information from auction rounds held in Brazil during the 2005–2017 period and employ multiple regression analyses to investigate the integrated effect of auction design elements, individual bids and project features on construction delays. We find that the schedule performance of wind power projects has been strongly improved with the introduction of a preliminary transmission capacity phase in auctions and when the risk of delays in the extension of the transmission grid was transferred to project developers. Projects contracted in auctions with longer lead times took more time to be completed but faced lower risk of project delays and penalties. Large scale plants were more prone to delays, while no significant difference was found between projects awarded in technology-specific or multi-technology auctions. Lower individual bids were not associated with longer delays, suggesting that apparently the Brazilian hybrid auction system was able to minimize the winner's curse. Our results add empirical evidence to the literature and may provide policy makers with new insights on how to design auctions to promote the expansion of renewable electricity supply.

1. Introduction

Auctions have become the main instrument to support electricity generation from renewable energy sources (RES) worldwide (del Río and Kiefer, 2021). According to IRENA (2019), 106 countries had adopted auction schemes to procure renewable-based electricity by the end of 2018 compared to only 8 by the end of 2005.

Brazil is one of the pioneers in the use of energy auctions to support the expansion of its electric power system. The Brazilian government structured its current auction programme in 2004 with the objectives of achieving energy security, improving the efficiency of electricity contracting for the regulated market, and promoting diversification of energy supply, especially for renewable energy sources (Tolmasquim et al., 2021). Since then, auctions have been the primary mechanism to procure electricity and capacity with a long-term focus. Thus, Brazil has one of the longest-running auction programmes in the world and it might offer important lessons on RES auction design and implementation (Pahle et al., 2021).

Auctions have the advantage that the remuneration level of generators is determined by a competitive procurement process that may lead

to falling remuneration levels and, consequently, ensure fair and cost-reflective tariffs. In addition, auctions offer more control over the expansion of electricity generation by specifying the desired capacity to be contracted and its implementation deadline (del Río and Linares, 2014).

Nevertheless, auctions are not able to overcome competitive market structure problems or mitigate all risks associated with the implementation of infrastructure projects (Tolmasquim et al., 2021). One of such risks is that projects might not be completed according to the estimated schedule. The delay of output caused by slippage in construction projects schedule imposes economic costs when the power system is short of capacity or supplying power from plants with high variable costs (Bacon et al., 1996). Renewable energy project delays and cancellations may lead to higher CO₂ emissions of the respective electricity market and to a reduced public acceptance of renewable energy (Bayer et al., 2018). Delays may also cause financial impacts on investors by increasing their financing charges and incurring on project loan repayments before revenues are generated (Bacon et al., 1996). Carvalho et al. (2020) studied the financial risk associated with the anticipation or delay in the completion of solar and wind projects in the

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Brazilian energy market, finding that delays significantly increase the probability of a failed business.

Thus, ensuring that power plants are built according to the contractual schedule is a legitimate concern for policy makers, reflected in several publications addressing mechanisms aimed at avoiding implementation delays in energy generation projects (del Río, 2017; Held et al., 2014; IRENA and CEM, 2015; Kreiss et al., 2017). Auction realization rates (i.e., the share of contracted capacity actually being commissioned) have been used in the literature as a proxy to quantify the effectiveness of RES auctions and investigate the effect of different auction design elements (Bayer et al., 2018; del Río and Linares, 2014; Matthäus, 2020; Shrimali et al., 2016; Winkler et al., 2018).

According to Quintana-Rojo et al. (2020), the effectiveness and efficiency of auction mechanisms is a topic that requires a greater load of econometric studies. Recent work has been published in the field (Anatolitis et al., 2022; Matthäus, 2020), but studies are typically limited to auction-level aggregated data (e.g., total deployed capacity, average bidding prices), and information regarding projects' construction status after auction deadlines is scarce. For example, the comprehensive online database developed by the European Union AURES¹ II Project gathers structured data about 820 auction rounds held in EU countries, but realization rates are available for only 6% of those (AURES II, 2022). This lack of structured project-level data prevents the assessment of other interesting dynamics related to the effective deployment of projects, such as the effects of individual bidding prices, geographical location, technology and scale.

Recently, few articles have employed econometric techniques on project-level data to assess the effects of RES auction design features. Probst et al. (2020) econometrically analyzed data on solar PV projects from auctions held in India between 2014 and 2017 to evaluate the effect of local content requirements (LCR) on average bid prices and project completion rates. They found that the LCR component has increased awarded prices significantly but had no effect on the realization rates of the projects. Based on data from onshore wind auctions conducted in Italy from 2012 to 2016, Cassetta et al. (2017) used a multiple linear model for an empirical analysis of project-related, firm-specific and auction design determinants of awarded base tariff. Nevertheless, the authors point out the difficulty of making a similar assessment on project delays and cancellation due to the lack of consistent data on the stages of construction of single awarded projects. Batz Liñeiro and Müsgens (2021) assessed project-specific, disaggregated individual bid data from the solar PV auction program in Germany. They employed regression analysis to explore the impact of project and developer characteristics, competition, and PV module prices on bid values and project realization.

In the Brazilian context, monthly reports on the expansion of electricity generation supply published by the Brazilian Power Regulatory Agency (ANEEL, 2021) contain data and statistics concerning project delays, but these are restricted to the execution of projects under construction. Based on expert interviews and extensive data analysis, Bayer et al. (2018) investigated the causes for delays in on-shore wind power plants from Brazilian auctions and discussed potential mitigation measures. We argue that a comprehensive assessment of the Brazilian experience with electricity auctions, from both renewable and non-renewable sources, with the employment of econometric techniques, might bring new quantitative insights on how design elements and individual project characteristics influence implementation schedules and auction effectiveness. Such analysis has the potential to bring important lessons regarding the future designs and planning of renewable energy auctions both in Brazil and elsewhere.

Therefore, the aim of this paper is to assess the implementation schedule performance of electricity generation projects contracted in auctions held in Brazil in the period between 2005 and 2017. Using a

database assembled by the authors gathering information published by regulators and energy market operators, we measure the incurred delays and perform statistical analyses seeking to identify possible correlations between observed delays and characteristics of projects (technology, scale, individual bids) and auction design elements (e.g., lead time and penalties).

Our work is structured as follows: in the next section we provide an overview of the Brazilian auction scheme, focusing on the main auction design elements adopted and on regulatory changes that have been implemented to deal with low realization rates. We also discuss the literature on project delays and RES auction design to identify elements assumed to be important for an empirical analysis on project delays. Section 3 details the methodology adopted, describing the database used, the definitions and analyses employed. In section 4 results are presented and their implications for renewable energy auctions planning are discussed. Finally, the last section brings the main conclusions and policy recommendations.

2. Background: literature review and electricity auctions in Brazil

An increasing number of countries uses auctions as an instrument to allocate support for renewable energies (IRENA, 2019). When designing their support schemes, policy makers must decide on several aspects that may impact cost-efficiency, social acceptability, level of competition, technology diversity, transaction costs, among others. The success of auctions in meeting policy objectives depends on the choice of design elements in specific settings, i.e. the devil is in the details (del Río et al., 2015).

The main objectives for policymakers in auction design are efficiency, in the sense of minimizing support level through low awarded auction prices (Anatolitis et al., 2022), and effectiveness, i.e., achieving high realization rates of awarded projects (Matthäus, 2020). In this paper, we focus on the latter, as the risk of delays and underbuilding of the auctioned projects have been a recurring concern (Azuela Elizondo et al., 2014; Bayer, 2018; del Río and Linares, 2014; Kreiss et al., 2017).

2.1. Electricity auctions in Brazil

The Brazilian auction system was created in 2004 with the objective of providing an efficient, transparent, and competitive instrument for the awarding of long-term PPAs for distribution companies, while reducing investor risks and facilitating project financing of new power plants. The auction system takes place in the context of the regulated market,² in which was traded about 64% of all electricity consumed in the country in January 2022 (CCEE, 2021a). These auctions are coordinated by the Ministry of Mines and Energy (MME) and performed by ANEEL and the Brazilian Power Trade Chamber (CCEE). In Brazil existing and new energies compete in separated processes, which allows the average price in the regulated market to be calculated apart from the marginal expansion cost, thus contributing to fair electricity rates (Tolmasquim et al., 2021). There are three types of electricity auctions related to the construction of greenfield projects:

New Energy Auction (LEN): Auction designed to contract electricity from greenfield projects, with the aim of meeting the increase in distribution companies' power demand, according to estimates provided by each of them prior to the auction round. These auctions can be held from

¹ (AUCTIONS for Renewable Energy Support II).

² In the Free Contracting Market (FCM), non-regulated agents, such as free consumers and energy traders, can directly negotiate bilateral contracts, whereby prices and contract terms are freely agreed on by the parties. However, purchase of electricity in the free market is exclusive for those end-users with installed load capacity equal or greater than 500 kW (ANEEL, 2021). Captive consumers are only allowed to buy electricity from the distributor in whose network they are connected, paying regulated rates.

three to six years before the starting delivery date. Power generation projects compete in the auction and each winner signs contracts with several distribution companies according to their expected increase in demand. Some generation projects, such as the Madeira River hydroelectric plants and the Belo Monte Hydroelectric Power Plant, because of their strategic nature and relevant public interest, are auctioned through a project-specific new energy auction called Structuring Project Auctions (LPE).

Reserve Energy Auction (LER): Auctions created to increase security in the electricity supply in the National Interconnected System by purchasing power from plants specially contracted for this purpose. Since renewable sources, mostly contracted in these auctions, are non-controllable sources and have inflexible dispatch, the concept of reserve energy is here interpreted as additional energy to the system that aims to reduce the depletion of hydroelectric power plant reservoirs (Tolmasquim et al., 2021). CCEE act as a single buyer and sign contracts with all the winners of the auction. As the reserve auction works as an insurance for the entire power system, the energy charges apply to all consumers from both the regulated and the free market.

Renewable energy auction (LFA): Auction designed to increase the share of renewable energy sources - which includes wind, solar, and small hydropower plants - in the Brazilian energy system. Despite the creation of a specific auction modality, so far only three rounds of LFA have taken place (in 2007, 2010 and 2015), and most RES capacity was procured through LEN and LER.

Table 1 summarizes the main elements in the design of the Brazilian auction scheme, classified according to the categories described in literature (del Río and Kiefer, 2021; IRENA and CEM, 2015). A more detailed description of Brazilian auctions regulatory and policy framework is beyond the scope of this paper and can be found elsewhere in literature (Correia et al., 2020; Tolmasquim et al., 2021).

Some design elements and contract obligations vary across distinct auction types and products, and some regulatory changes have been implemented over the years. The adoption of different design elements for each auction round considered in this study is shown in Table A1. In the next paragraphs, we discuss how some of those variations in auction design and individual project features might have affected implementation delays and formulate predictions to be empirically tested in our study.

2.2. Penalties and exposure to spot market prices

In case of non-realization or delay, penalties include obligation to reimburse the energy that was not delivered, execution of financial guarantees, and further sanctions, such as termination of the contract with contractual fees, a ban from auctions, or administrative fines up to 10% of the respective investment. The legal consequences of each individual case are defined by the regulatory authorities, who can make an ex-post decision on which administrative measure is the most suitable in order to improve the implementation process (Bayer et al., 2018). The specific reimbursement obligation in case of delays depends on the auction type. For projects awarded in Reserve Energy Auctions (LER), a charge on the non-supplied electricity is applied based on project bid price, while in other auction types project developers must purchase the missing quantity on the electricity market, bearing the risk of market price volatility if the market price exceeds the contractual remuneration. According to (Bayer et al., 2018), the obligation to bear market price risks creates an even stronger financial incentive for on-time implementation, since spot market prices have been historically higher than the auction price, especially in dry seasons. Following these arguments, we test the prediction that projects awarded in LER auctions are subject to longer delays.

2.3. Technological banding

Each auction round includes one or more products, auctioned in

Table 1

Brazilian auction scheme basic design elements

Source: Own elaboration based on (Bayer et al., 2018; Tolmasquim et al., 2021).

Metrics for volume setting and disclosure	The contracted volume is set in terms of generation (MWh)
Timing (schedule)	There is no fixed schedule of auctions, but these are organized on a regular basis – usually 2 rounds per year.
Technological Diversity	Technology-specific (TS) or multi-technology (MT) auctions.
Geographical Diversity	Geographically neutral - no requirement to deploy the project in a given location (except for project-specific auctions).
Pricing rule ^(a)	Hybrid model in two phases: 1 - Descending-clock auction with uniform price, starting with a ceiling price defined by regulators. Bidders indicate how much they are willing to supply at this price. The auctioneer then lowers the price until the desired supply level is met, plus a certain margin, to stimulate competition in the second phase. 2 Final pay-as-bid round for the winners of first phase. The lowest (sealed) bids are awarded PPAs at the price of their bid
Winner selection criteria	Price-only
Remuneration	Full payment through long-term PPAs, signed with either CCEE (Reserve auctions) or a pool of distribution companies
Lead time	From 2 to 6 years
Pre-qualifications	
Physical	Bidders must secure social and environmental permits, land use rights and interconnection agreements to be allowed to register for bidding.
Financial	- Bid bond (1% of total estimated investment costs) before joining the bidding stage - Winners must provide completion bonds (5% of estimated investment) before contract signing - Project developer financial health (General liquidity indicator threshold to participate).
Penalties	In case of delays, project developers are subject to administrative fines, obligation to reimburse energy costs, termination of contract, or exclusion from future auctions.
Risks	
Seller	Construction, operation, equipment performance risks and exposure to the spot market (in the case of 'quantity contract' modality ^(b))
Buyer	Inflation and exposure to the spot market, in the case of the 'availability contract' modality

^a From 2017 onwards, the bidding process changed to a continuous trade reverse auction in two phases. In the first one, 'temporary winners' are selected through sealed bids with a price and quantity. The second phase consists of descending clock iterations in which any temporarily disqualified bidder can replace a temporary winner by submitting a bid lower than the marginal price minus a decrement defined by the auctioneer.

^b The PPAs awarded to auction winners may be of two types: quantity and availability contracts. In the quantity modality, the seller is responsible for delivering the contracted amount of electricity, assuming the operating costs and the risks of variations in energy generation and short-term market prices. In the availability contracts, generators receive a fixed revenue for offering their plant's available capacity, while buyers bear variable costs (fuel) and generation imbalance settlement costs/revenues.

separate tenders, which can be either technology-specific or multi-technology (i.e., where different technologies compete directly). The question of technological banding in auctions is a much debated topic (Lena Kitzing et al., 2019; Lucas et al., 2020). While multi-technology auctions have the potential to maximize cost-efficiency (Kreiss et al., 2021), technology-specific designs may stimulate technological learning and avoid crowding out of less mature technologies (Lena Kitzing et al., 2019). Also, technological banding allow support levels and auction rules to be adjusted according to market conditions and technology maturity (Polzin et al., 2015), which may potentially increase realization rates. However (Matthäus, 2020), analyzed RES auction rounds

worldwide and found no empirical evidence that technological banding affect effectiveness. In this study, we test whether, in the Brazilian context, projects awarded in technology-specific auctions are less prone to delays.

2.4. Lead time

According to IRENA and CEM (2015), lead time is a key attribute of electricity commercialization auctions, being typically defined based on expectations regarding the time required for the construction of projects and execution of the administrative procedures related to them. If the lead time is shorter than necessary, entrepreneurs have little margin for error, resulting in a higher risk of delays that can result in penalties. However, auctions held too far in advance can lead to a certain degree of speculation, attracting, for example, investors who plan to delay the start of construction in anticipation of possible reductions in development costs. Sovacool et al. (2014) consider that longer construction lead-times deal with several uncertainties during the construction process, including unforeseen changes in interest rates, availability of materials, exchange rates, severe weather and labor strikes, making planning and financing difficult.

In the Brazilian case, investors have the option to sell in the free market whatever energy is generated before the auctioned PPA's supply period, which might create an incentive to anticipate the commercial operation of projects contracted in auctions with longer lead times. However, this is not a straightforward decision for investors, depending on several factors including bidding prices, market conditions and expectations. Dalbem et al. (2014) modeled bidders' price decision in Brazilian A-5³ wind auctions as real options and concluded that given low bid prices, winners might be tempted to defer the start of construction, expecting more favorable equipment and energy prices, though evidence from European RES auctions (Anatolitis et al., 2022) suggests that this behavior might be inhibited when prequalification criteria and penalties are in place.

Based on this debate, we test the hypothesis that longer auction lead times increase the on-time implementation of awarded projects.

2.5. Transmission related delays

In Brazil, transmission was traditionally planned, auctioned and built after generation auction winners were revealed and their location and nature were defined. However, in practice, delays in the implementation schedule of transmission facilities, related to environmental constraints or underbidding in transmission auctions, have resulted in many cases in which generation facilities are ready to operate by the time their contractual delivery date is achieved, but the output of renewable generators cannot reach the market because transmission capacity reinforcements are not ready in time (IRENA and CEM, 2015).

Herrera et al. (2019) point to the transmission capacity deficit as the main bottleneck to the expansion of generation in the Northeast Region, which concentrates most of the Brazilian potential for wind and solar sources. Indeed, a significant part of the delays incurred in the generation projects located in this region did not result from situations related to the implementation of these projects themselves, but from delays in the associated transmission projects. This problem is not unique to Brazil: delayed connection to the grid has been identified as one of the key factors accounting for the gap between the technical potential and the actual output of Chinese wind farms (Huenteler et al., 2018; Lu et al., 2016; Zhang et al., 2013).

In some of Brazil's early auctions, such transmission risks were allocated almost entirely to electricity buyers. The PPAs awarded as a result of the auctions included a clause that exempted the project developers from their contractual obligations and guaranteed

remuneration if the project developer could prove that their plant was ready to operate, while buyers had to arrange alternative procurement options. Since 2013, such waiver has been removed, and the risk of the unavailability of transmission capacity has been fully transferred to generation project developers, who became obligated to fulfill their contractual obligation to supply electricity even in case of delays in the extension of the transmission grid.

Another regulatory change implemented for dealing with this problem was the introduction, for certain electricity auctions with shorter lead times, of a preliminary phase to select the projects with connection feasibility through competition among projects with the same connection point. In this preliminary phase, bidders submit a single bid with price and quantity for each project, and bids are classified by their price at each connection point. Projects that exceed the 'transmission margin'⁴ are excluded from the auction, and winners then engage in the general auction process (Tolmasquim et al., 2021).

Bayer et al. (2018) observed a reduction trend on wind power project delays due to transmission grid issues after the implementation of both auction design amendments. Based on this evidence, and including outcomes from recent auction rounds, we empirically test whether each of those regulatory changes effectively reduced project delays.

2.6. Pricing rule, underbidding, and underbuilding

As described in Table 1 - Brazilian auction scheme basic design elements, the regular Brazilian auction procedure is a hybrid model that occurs in two phases: a first descending clock phase with uniform pricing, followed by a pay-as-bid final round in which the lowest (sealed) bids are awarded PPAs at the price of their bid. The auctioneer may expect lower prices under a pay-as-bid sealed auction, but the trade-off might be a higher risk of the winner's curse in this scenario (Tolmasquim et al., 2021). Haufe et al. (2018) points out the risk of irrational underbidding under pay-as-bid models: bidders may tend to bid lower than their costs, leading to project delays and cancellations. On the other hand (Matthäus, 2020), found no empirical evidence that the pricing rule affect effectiveness. Based on this debate, we formulate the hypothesis that, among projects awarded in an auction round, individual projects with lower bid prices are more prone to delays.

2.7. Project technology and scale

There is a panoply of studies assessing the magnitude and frequency of delays and showing it is a pervasive issue in infrastructure projects worldwide. Sovacool et al. (2014) assessed construction costs and schedule affiliated with 401 power infrastructure projects worldwide, finding that electricity infrastructure is prone to delays and cost overrun issues almost independently of technology or location, though wind, solar and small thermoelectric power plants showed significantly lower risks of cost overruns and construction delays in comparison to nuclear (Portugal-Pereira et al., 2018) and large hydropower plants. Bacon and Besant-Jones (1998) show that large projects have larger cost and time overruns even in proportion to their size. Callegari et al. (2018) suggested that solar and wind sources present a delivery time shorter than initially estimated due to their rapid learning curves. Other studies for Brazil have arrived at similar results (Köberle et al., 2018). However, studies addressing project delays in the first rounds of wind power auctions in Brazil have identified low rates of on-time implementation, with many projects being delayed for more than one year (Azuela Elizondo et al., 2014; Bayer, 2018).

Biomass-powered thermoelectric plants are expected to face lower delays in the recent Brazilian auctions, since many of those are actually

³ Auctions for electricity with first delivery date 5 years after the auction.

⁴ Limiting draining capacity calculated by the Energy Research Company (Empresa de Pesquisa Energética, or EPE) and the Independent System Operator (ONS) for each connection point indicated by the registered projects.

brownfield processes associated to sugarcane mills, initially built to generate electricity to supply their own consumption and then retrofitted to sell the surplus power. Moreover, sugarcane mills are usually near the load centers, thus implying reduction of transmission grid connection issues (Watanabe et al., 2020).

In an attempt to address some of these questions in the context of the Brazilian auction program, we include in our analysis variables to account for the effect of project scale (measured by installed capacity) and technology.

3. Methodology and data

3.1. Research design

Initially, following the procedure described in the next subsection, we build a database containing information on several project, auction, bids, and PPA features.

We then employ a multiple regression technique to test the hypotheses discussed in Section 2 on the influence of auction design elements, underbidding, project scale and technology on measured delays. This approach allows for inference on the integrated effect of several variables on the observed delays, which might not be captured in a descriptive analysis of the database. If statistically significant⁵ correlations can be established, then quantitative knowledge about those effects might bring insights to the improvement of auction mechanisms.

3.2. Database

The database used in this study (Diniz, 2022) was compiled gathering information published by the Brazilian Power Trade Chamber (CCEE), the Brazilian Electricity Regulatory Agency (ANEEL) and the Independent System Operator (ONS). Basic information on project characteristics, commercial operation date, and main PPA features were obtained from the combination of two CCEE's monthly updated publications: the "Consolidated Auction Results" (CCEE, 2021a) and the "Market Information Bulletin - Individual Project Data" (CCEE, 2021b).

Additionally, we accessed auction notices and PPA drafts to verify, for each case, the auction design features employed, such as technological banding (technology-specific or multi-technology products), preliminary phase for project elimination by connection point and allocation of transmission delays risks to project developers.

The resulting database contains, for each project, information regarding energy source, geographic location, installed capacity; date of effective start of commercial operation, project status (in operation, delayed or cancelled); auction type, date and relevant design features; contract modality (quantity or availability); start and end date of the contracted supply period; bidding price and average energy traded per year.

We focus on auction rounds related to the construction of new power plants⁶ and for which deadlines had been met by December 2021. We collected information on 1068 generation projects that sold a total amount of 6682 TWh in 37 auction rounds, held between 2005 and 2017. The database comprises wind power plants, hydroelectric power plants, small hydropower plants (<30 MW), photovoltaic solar plants (solar PV), and thermal power plants fueled with natural gas, biomass, oil and coal.

⁵ Unless otherwise stated, this paper adopts a general significance level of 0.05.

⁶ Auctions intended for contracting energy generated by plants already built and in operation (Existing Energy Auctions and Adjustment Auctions) were not included in the base.

Regarding wind power and solar photovoltaic plants, due to tax incentives and regulatory issues,⁷ it is usual that larger generation complexes are virtually segmented into smaller adjacent farms. For this study, based on information extracted from ONS (2022), 660 projects were grouped into 130 complexes, each considered as a single project, as they usually share the same facilities, developers, licensing and construction process. For projects that sold energy in more than one auction, we considered only the first auction in which the project was awarded a contract. In the end, 523 observations remained in the full database,⁸ which comprises all bid winners, including projects that have been officially cancelled or those with extremely large delays and low probability of realization.

For the econometric analysis developed in this study, we restricted the sample in 4 steps. First, we dropped 83 fossil-fuel thermal plants and kept just the most relevant sources in terms of number of projects and installed capacity: Wind, Solar PV, Biomass and Hydropower (large and small) plants. Second, we excluded 50 officially cancelled projects. Third, we removed 28 projects with extremely small implementation times (less than one year), which does not seem compatible with construction times for the technologies. Finally, we filtered out 23 projects that have been either delayed or anticipated by more than 3 years from the auction deadline, which indicates high probability that the schedule of such projects is related to specificities beyond the project features and auction design elements under analysis. Thus, the final sample for statistical analysis contains 339 projects, with 45 GW total installed capacity and that sold 3802 TWh in 34 auction rounds.

3.2.1. Dependent variables

Our main dependent variable is project delay, defined as the difference between the auction completion deadline and the commercial operation date, which corresponds to the moment when the first generating unit of the plant was connected to the power system. Delays are expressed in absolute terms (days), and negative values mean that commercial operation started before the beginning of the contractual electricity supply period. For power plants that did not start operation by the end of 2021, delay is calculated as the time elapsed between auction deadline and the reference date of this study (Dec. 21). We choose delay as our main variable of study because it measures how implementation schedules deviate from the targets agreed by the auctioneer and project developers, which may jeopardize projects feasibility and RES auction effectiveness. Alternatively, we also employ as dependent variable *Realization Time*, which is defined as the time elapsed between auction date and project commercial operation date.

Table 2 summarizes our potential explanatory variables, chosen according to the hypothesis and predictions discussed in Section 2, while summary descriptive statistics of the final sample used in the analysis is presented in Table 3.

3.3. Model selection and implementation

To quantify the integrated effect of potential explanatory variables in project delays, we estimate the following multi-variate model:

$$\text{Delay}_i \sim \beta_0 + \beta_1 \cdot \text{Predictors}_i + \beta_2 \cdot \text{Controls}_i + \varepsilon_i \quad (3)$$

In this model, *Predictors* is a vector including the potential explanatory variables and some relevant interactions between them, while *Controls* is a vector of control variables and covariates introduced in each model. *Auction Year* corresponds to a covariate representing the year when the auction round took place to capture possible time-varying

⁷ Power plants from renewable energy sources with less than 30 MW of installed capacity were eligible for reduced tax rates and discounts on transmission and distribution system fees.

⁸ The full database (Diniz, 2022) includes raw data and a version in which projects are not grouped into complexes.

Table 2
Description of potential explanatory variables.

Variable	Type	Description
<i>Technology_Specific</i> ⁽¹⁾	Dummy	1: awarded project did not compete directly with other technologies; 0: project awarded in a multi-technology auction.
<i>Lead_Time</i>	Continuous	Period (in years) between auction date and the start of supply period defined in the PPA.
<i>Dev_Transmission_Risks</i>	Dummy	1: transmission grid connection risks are allocated to generation project developer; 0: risk of unavailability of transmission capacity is transferred to buyers.
<i>Preliminary_Phase</i>	Dummy	1: project awarded in an auction with a preliminary phase to shortlist projects that fit in the available transmission capacity; 0: project awarded in auction without transmission capacity phase.
<i>LER</i>	Dummy	1: project awarded in a Reserve Energy Auction type (LER), 0 otherwise.
<i>Individual_Bid_Ratio</i>	Continuous	Ratio between the individual project bid price and the auction round average price.
<i>Wind</i>	Dummy	1: on-shore wind power plant; 0 otherwise.
<i>Biomass</i>	Dummy	1: biomass-powered thermoelectric plant; 0 otherwise.
<i>Hydro</i>	Dummy	1: Hydropower plant (large or small); 0 otherwise.
<i>Proj_Installed_Capacity</i> ⁽²⁾	Continuous	Project installed capacity (in MW).

(1) Auctions with competition between large and small hydro, or between different thermal sources (e.g. gas, coal, biomass), are considered as multi-technology auctions. (2) Logarithmic transformation applied to normalize the residuals of the model, which would otherwise be positively skewed.

Table 3
Summary descriptive statistics of final sample.

	Mean	St. Dev.	Min	Median	Max
Delay	70	416	−1092	59	1086
Realization_Time	1209	458	474	1152	2559
Lead_Time	3.120	0.967	1.353	2.970	4.704
Individual_Bid_Ratio	1.009	0.131	0.598	1.001	1.553
Proj_Installed_Capacity	133	667	0.56	45	11233
ln_Proj_Installed_Capacity	3.743	1.304	−0.580	3.807	9.327
Dev_Transmission_Risks	0.593	0.492	0	1	1
Preliminary_Phase	0.124	0.330	0	0	1
Technology_Specific	0.522	0.500	0	1	1
Wind	0.327	0.470	0	0	1
Solar_PV	0.112	0.316	0	0	1
Biomass	0.183	0.387	0	0	1
Hydro	0.351	0.478	0	0	1
Quantity_Contract	0.649	0.478	0	1	1

Note: N = 339 observations for all variables.

hidden effects, such as technological development, the learning curves of renewable energies, and experience acquired by auctioneers and project developers. It is added as a continuous variable to avoid multicollinearity issues arising from splitting the study period into bins that might be correlated with the introduction of some design elements under analysis. In order to account for possible effects of the two different PPA modalities described in Table 1, we include *Quantity_Contract* as a dummy control variable that indicates whether the PPA is a Quantity (=1) or Availability (=0) contract.

We conducted several tests⁹ with fitted models to check for the assumptions of normality of residuals, multi-collinearity and heteroskedasticity that might bias the inference of our results. While other

assumptions for a usual Ordinary Least Square (OLS) regression have been met, we concluded that there is heteroskedasticity in our data. Consequently, we follow Wooldridge (2018) and report heteroskedasticity-robust standard errors and test statistics for OLS.

All computations were conducted in R version 4.2.1 (R Core Team, 2022), and we used packages “performance” (Lüdtke et al., 2021) and “lmtest” (Zeileis and Hothorn, 2002) for calculations, tests and model diagnostics. Additionally, we used packages “stargazer” (Hlavac, 2022) and “flextable” (Gohel and Skintzos, 2022) for creating tables, and “ggplot2” (Wickham, 2016) for plots.

4. Results and discussion

In this section, we test whether the predictions from literature discussed in Section 2 hold for our empirical data. Table 4 summarizes the results of our models.

Our baseline model (Model 1) focuses on the potential explanatory variables described in Table 2. Given the historical prominence of grid connection issues in wind power project delays in Brazil (Bayer et al., 2018), we include interaction terms to account for the effect of regulatory changes intended to mitigate transmission-related delays (*Dev_Transmission_Risks* and *Preliminary_Phase*) on on-shore wind projects. For Solar_PV projects, we add only the interaction with *Preliminary_Phase* since *Dev_Transmission_Risks* equals one for every awarded Solar PV project. In Model 2, we add the control variable *Quantity_Contract* to account for potential effect of different PPA types (Quantity or Availability contract modalities). Furthermore, in Model 3 we add the *Auction_Year* covariate to account for continuous time-varying effects. Finally, in Model 4 we replicate all explanatory and control variables from Model 3, but *Realization_Time* is adopted as the dependent variable. The estimators in our full model are able to explain a fair portion of the variation in project delays (Model 3, adjusted R² of 0.25), and even more in *Realization_Time* (Model 4, adjusted R² of 0.38).

The coefficient on *Technology_Specific* is insignificant in all models, which suggests that competing in specific or multi-technology auctions did not play a major role in steering the implementation schedule of power plants. This result adds to the evidence from the worldwide RES auctions assessment performed by Matthäus (2020), who found no association of technological banding with the realization rate of auctions.

Lead_Time has a significant positive effect on *Realization_Time* in Model 4, but the estimated coefficient (245 days for one additional year in *Lead_Time*) suggests that the increase in project implementation periods is shorter than the auction lead time extension. Accordingly, the effect of *Lead_Time* on the delays of awarded projects is significant and negative on Models 1, 2 and 3. This is an expected result since with longer realization periods project developers are granted enough time to comply with necessary administrative requirements and complete the power plants. Also, for generators whose grid access is conditioned to grid expansion, longer deadlines may provide the time required to the execution of grid expansion activities (IRENA and CEM, 2015). Although shorter periods might be desirable for auctioneers willing to meet electricity demand and RES expansion targets in the short term, literature suggests that the reduced risk of delays and penalties faced by project developers in auctions with longer lead times might lead to lower bid prices (del Río and Linares, 2014; Hochberg and Poudineh, 2018; IRENA and CEM, 2015). Anatolitis et al. (2022) empirically found that, in European RES auctions, longer realization periods were associated with lower awarded prices, though this effect is reversed when long lead times are adopted in combination with financial prequalification. Similar analyses on the effect of lead time on bid prices and its interrelations with other design features, both in Brazil and elsewhere, may bring additional insights to policy makers designing RES auctions.

The coefficients on LER are insignificant in all models. In LER auctions, penalties are calculated based on awarded PPA prices, while in other auction types project developers are exposed to market price volatility risks in case of delays. Apparently, adding this risk to the

⁹ Shapiro-Wilk test for normality of residuals, Breush-Pagan test for heteroskedasticity and Variance Inflation Factor (VIF) for multicollinearity.

Table 4
Regression results.

	Delay			Realization_Time
	(1)	(2)	(3)	(4)
Technology_Specific	65.42 (65.99)	63.32 (66.49)	142.08 (78.18)	142.08 (75.94)
Lead_Time	−98.39*** (29.03)	−95.97** (29.46)	−119.77*** (28.78)	245.23*** (31.38)
LER	−0.74 (61.20)	20.37 (82.50)	−100.55 (107.07)	−100.55 (100.05)
Individual_Bid_Ratio	−10.02 (189.22)	−7.43 (189.21)	34.31 (185.98)	34.31 (168.25)
Solar_PV	97.82 (92.02)	85.06 (99.06)	23.09 (101.82)	23.09 (114.69)
Wind	285.45*** (69.28)	263.62** (91.82)	339.88*** (102.17)	339.88*** (92.01)
Biomass	−172.90* (78.72)	−211.59 (125.11)	−83.33 (147.97)	−83.33 (130.95)
In_Proj_Installed_Capacity	46.90** (16.86)	48.24** (17.19)	64.20*** (17.25)	64.20*** (18.42)
Dev_Transmission_Risks:Wind	−367.07*** (67.93)	−378.70*** (68.22)	−416.98*** (73.21)	−416.98*** (89.87)
Preliminary_Phase:Wind	−239.70** (78.41)	−229.33** (81.59)	−347.64*** (97.47)	−347.64*** (126.77)
Preliminary_Phase:Solar_PV	−78.24 (101.04)	−85.23 (101.50)	−100.91 (105.65)	−100.91 (123.62)
Constant	158.95 (233.81)	177.47 (238.88)	64.76 (237.89)	64.76 (227.55)
Auction Year Covariate	No	No	Yes	Yes
Contract Type Control	No	Yes	Yes	Yes
Observations	339	339	339	339
R ²	0.26	0.26	0.28	0.40
Adjusted R ²	0.24	0.24	0.25	0.38

*p < 0.05.

**p < 0.01.

***p < 0.001.

existing prequalification criteria and penalties did not create relevant further incentive to the on-implementation of projects.

Individual Bid Ratio was not significantly related to delays incurred by the contracted power plants, which means that apparently there was not a problem with project delays caused by excessively low bids of individual projects. This result might be associated with the hybrid system adopted in Brazilian auctions in which, according to [Del Río and Linares \(2014\)](#), the first descending-clock phase allow for price discovery and minimizes the winner's curse, while the second sealed-bid phase prevents collusion.

Biomass-powered thermoelectric plants were less prone to delays, which is in line with prediction from Section 2, though the coefficient on *Biomass* is only significant in Model 1. On the other hand, *Wind* showed a significantly positive coefficient in all models, which reflects the poor on-time implementation of on-shore wind farms in Brazilian early auctions (see [Fig. 1](#)), mainly caused by grid connection issues ([Bayer et al., 2018](#)).

The coefficients on the interaction term *Dev. Transmission Risks:Wind* are negative and significant in all models, indicating that the allocation of transmission risks to generators is associated with a strong improvement in the implementation schedule of wind power plants. This corroborates [IRENA and CEM \(2015\)](#), who point out that this design option is an efficient way of improving the on-time realization rate, since it forces generation developers to prioritize the development of projects in sites with lower risk of network connection, but bidders might include this liability as a risk premium in their bids, potentially resulting in price increases. Although wind power bid prices have indeed increased after this change in auction rules, further studies are needed to investigate eventual causal relations, since such increase may be associated to a plethora of factors lying outside tender rules, such as changes on Brazilian currency (BRL) exchange rates and the rise of infrastructure financing interest rates. The effect of the allocation of grid connection risks on wind power plants delays can be seen in [Fig. 2](#), which shows the distribution of project delays for different technologies in our sample.

Regarding the adoption of a preliminary transmission capacity stage, the terms *Preliminary_Phase:Wind* and *Preliminary_Phase:Solar_PV* showed a negative effect on awarded project delays, though for solar photovoltaic projects this reduction is not significant at usual levels. This result indicates the success of this design element intended to reduce grid connection issues, especially for on-shore wind power plants, for which grid connection issues might have been more relevant to completion schedules than for other RE sources, such as Solar PV.

Proj.Installed.Capacity has a significant positive effect on project delays, which is in line with previous findings in the literature ([Callegari et al., 2018](#); [Portugal-Pereira et al., 2018](#); [Sovacool et al., 2014](#)), that large scale projects are more prone to cost overruns and schedule slippages, due to their inherently higher complexity and more associated uncertainties. Thus, although this finding may advocate towards the adoption of auction design mechanisms that favor small-scale projects, which might lead to higher on-time implementation rates, restricting the participation of larger projects can impede economies of scale and reduce competition levels in the short-term, leading to lower support cost efficiency. For example, [Anatolitis et al. \(2022\)](#) empirically found a price-increasing effect in RES auctions implemented exclusively for small-scale (<1 MW) projects. On this issue, the literature ([Flyvbjerg, 2014](#)) does not always recommend avoiding larger projects. Instead, it recommends having a very careful pre-FID (final investment decision) analysis of them, raising all risks that might be incurred with larger projects (e.g. cost inflation and delays) and addressing the possible adaptations that might be needed during the implementation phase.

It is important to note that, although multiple regression techniques allow controlling for certain confounding variables, it is not possible to determine from data the presence of unmeasured lurking variables that might be affecting the observed delays. For this reason, in the analyses employed in this study, the association between variables and outcomes may bring interesting insights but does not necessarily imply the existence of causal relations.

Another limitation of our study is the potential bias introduced by the exclusion from the sample of projects delayed by more than 3 years or that have been ultimately cancelled. Although some variables from our study may have influence on project cancellations, our sample size and information available do not allow for establishing robust inferences. Moreover, cancellations shall be analyzed in each case, since it may be affected by a multitude of specific factors and circumstances not considered in our analysis, including individual negotiation with regulators on the application of potential sanctions, contract termination

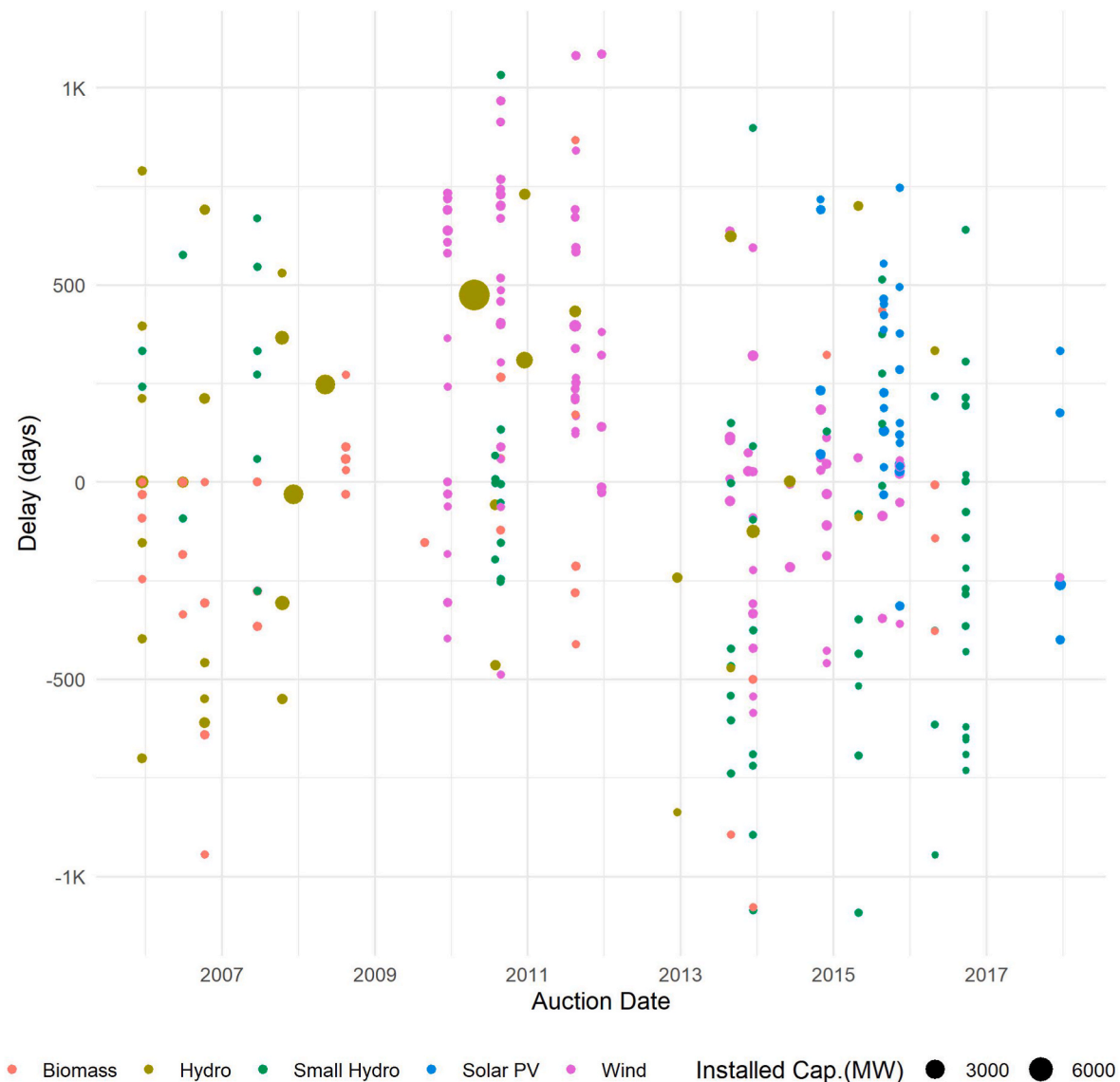


Fig. 1. Project delays per technology, scale and auction date. Source: Own depiction based on (Diniz, 2022).

opportunities due to electricity oversupply, land-use conflicts, problems with suppliers, and the effect of economic cycles¹⁰.

5. Conclusions and policy implications

Brazil has a long-running experience with auctions as an instrument to procure electricity both from renewable and non-renewable sources. We argue that important lessons on policy options for RES auction design may be drawn from the Brazilian historical experience on the implementation schedule of awarded projects.

We built and published a unique database containing detailed information on 522 generation projects with a total installed capacity of 74.5 GW, which sold electricity equivalent to 6682 TWh in 37 auction

rounds held in Brazil between 2005 and 2017. From this universe, we selected a sample of projects from the sources that had a considerable expansion in Brazil in the period (Wind, Solar PV, Biomass and Hydro-power) and employed multiple regression analyses to investigate the integrated effect of auction design elements and individual bids and project features on observed project delays.

Our results add empirical evidence to the literature and may provide policy makers with new insights on how to design effective and efficient auctions to promote the expansion of renewable electricity supply.

We find that biomass-powered thermal plants were less prone to delays than other types of power plants, while on-shore wind projects had significantly longer delays in comparison to other sources. However, we present empirical evidence that the on-time implementation performance of wind power projects has been strongly improved after the introduction of regulatory changes aimed at mitigating problems related to the connection of power plants to the transmission grid, which was the key determinant for project delays. This includes, first, the assignment to project developers of the risk of delay in the transmission grid extension; then, the introduction, in auctions with shorter lead times, of a preliminary phase to shortlist projects that fit in the available transmission capacity in each connection point. The Brazilian experience shows that policy makers might consider the adoption of similar auction

¹⁰ For example, there was a 36% devaluation of the Brazilian currency (BRL) against the US Dollar in the year following the first auction round with participation of photovoltaic technology (6th LER, 2014). Contracted projects faced a significant challenge, with escalation of costs for imported equipment that had to be purchased at an exchange rate higher than estimated at bidding time (Barbosa et al., 2020). From 11 solar PV plants awarded in that auction round, 6 have been cancelled, while the others were completed out of schedule.

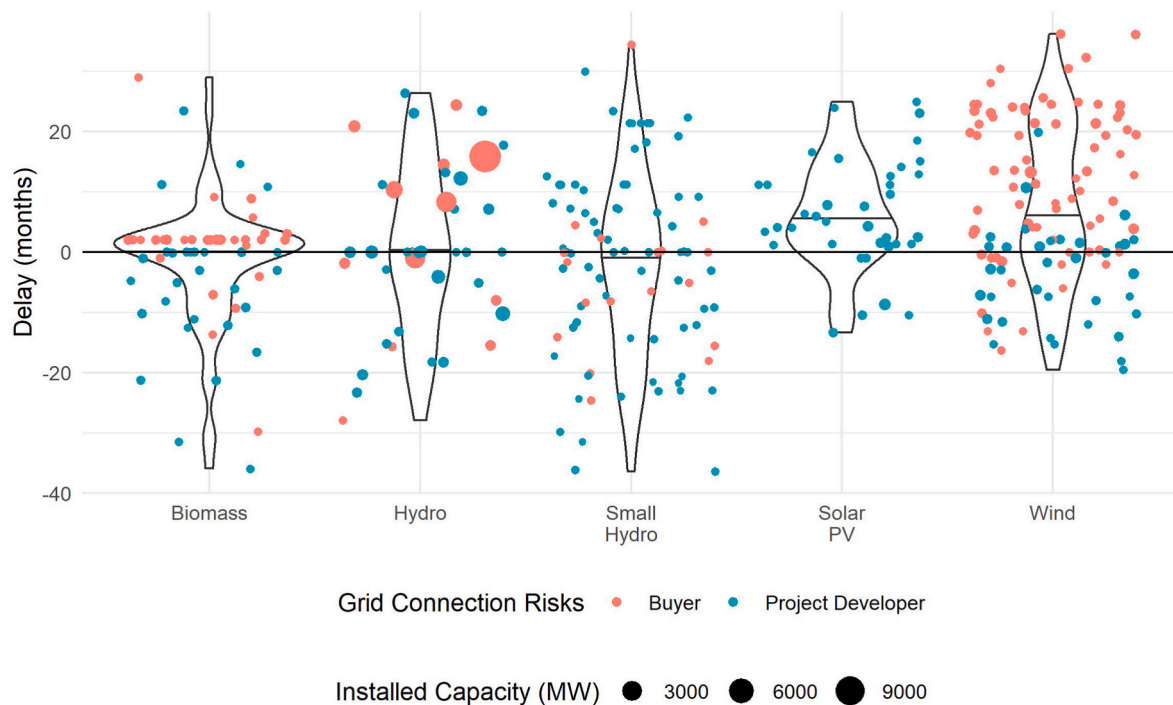


Fig. 2. Project delays by project technology and allocation of grid connection risks.

design mechanisms that may extensively reduce project delays, especially in countries facing the challenge of transmission grid expansion to connect new renewable energy capacity.

Projects contracted in auctions with longer lead times took more time to be completed but were less prone to delays. Thus, extending the period available for the construction of power plants might be an option to reduce project developers' risks of incurring in penalties, which in theory may lead to lower bid prices.

We find no significant difference in terms of completion schedule performance between projects awarded in technology-specific auctions and those that competed directly with other technologies in the same tender. This is in line with findings from [Matthäus \(2020\)](#), that technological banding do not have a major impact on RES auction realization rates. Therefore, auctioneers' decision on whether to adopt this design element should focus on other policy objectives, such as fostering immature technologies, diversifying the generation mix or minimizing support cost levels.

Moreover, no association was found between low bids of individual projects and the delays incurred by the respective power plants, which apparently supports the theoretical prediction that hybrid auction systems as the one adopted in Brazilian auctions may minimize the winner's curse.

Large scale projects were significantly associated with longer delays, which corroborates results from previous studies on electricity infrastructure worldwide. This suggests that policy makers concerned with on-time implementation rates should consider the adoption of auction design mechanisms that favor small-scale projects. However, such decision must carefully consider the risks and benefits in terms of competition, economies of scale and support cost efficiency, as well as other policy objectives such as actor diversity, social acceptability and decentralization of renewable energy production.

The conclusions from this work and the published database open the door for insightful further research. First, adding information on projects shareholder structure might enable the assessment of how developer size and experience impact project delays. Likewise, data on transmission auctions and project schedules, in combination with the generation

projects database from this study, could eventually provide insights into potential improvements on the integrated expansion of generation and transmission infrastructure. Also, similar studies on the experience of other countries may bring different perspectives and solutions. Finally, it would be interesting to include other relevant variables in the database and apply econometric techniques to analyze the effect of the design elements discussed in this paper on bid prices and other policy objectives, which might shed light on interrelations and trade-offs between them.

CRediT authorship contribution statement

Bruno Andrade Diniz: Conceptualization, Methodology, Writing – original draft, Writing – review & editing. **Alexandre Szklo:** Conceptualization, Methodology, Supervision, Writing – review & editing. **Maurício T. Tolmasquim:** Conceptualization, Methodology, Supervision, Writing – review & editing. **Roberto Schaeffer:** Conceptualization, Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

We have shared our research data on Mendeley Data Repository

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Appendix A

Table A.1

Auction rounds for which deadlines have been met

Auction Type	Official name	Auction date	Auction with preliminary transmission capacity stage	Transmission risks allocated to generators	Auction products				Awarded projects	
					Technological banding	Technologies	Contract Type	Lead Time (months)	Number of projects	Capacity (MW)
LEN	01° LEN	2005-12-16	No	Yes	Specific	Small Hydro	Quantity	25	2	739
								37	8	1412
								49	1	84
								25	8	1604
LEN	02° LEN	2006-06-29	No	Yes	Multi-technology	Thermal (any)	Availability	37	10	945
								49	2	566
								31	10	587
								31	14	1280
LEN	03° LEN	2006-10-10	No	Yes	Multi-technology	Small Hydro, Large Hydro	Quantity	51	5	891
LFA	01° LFA	2007-06-18	No	Yes	Specific	Small Hydro	Availability	51	10	1948
							Quantity	31	6	107
LEN	04° LEN	2007-07-26	No	Yes	Multi-technology	Thermal (any)	Availability	31	8	502
								31	8	502
LEN	05° LEN	2007-10-16	No	Yes	Multi-technology	Small Hydro, Large Hydro	Quantity	31	12	1808
								31	12	1808
LPE	UHE Santo Antônio	2007-12-10	No	No	Specific	Large Hydro	Quantity	51	5	2140
								49	1	3151
LPE	UHE Jirau	2008-05-09	No	No	Specific	Large Hydro	Quantity	57	1	3300
LER	01° LER	2008-08-14	No	No	Specific	Biomass	Availability	5	2	176
LEN	06° LEN	2008-09-17	No	Yes	Multi-technology	Thermal (any), Small Hydro	Availability	17	29	2000
								28	10	1935
LEN	07° LEN	2008-09-30	No	Yes	Multi-technology	Wind, Small Hydro, Thermal (any)	Availability	52	22	5096
								52	1	350
LEN	08° LEN	2009-08-27	No	Yes	Multi-technology	Large Hydro	Quantity	29	1	68
								29	1	68
LER	02° LER	2009-12-14	No	No	Specific	Wind	Quantity	31	27	1744
LPE	UHE Belo Monte	2010-04-20	No	No	Specific	Large Hydro	Quantity	57	1	11233
LEN	10° LEN	2010-07-30	No	No	Multi-technology	Large Hydro, Small Hydro	Quantity	54	7	823
LER	03° LER	2010-08-25	No	No	Specific	Biomass	Quantity	4	3	228
								16	1	70
								29	1	80
								37	11	548
LFA	02° LFA	2010-08-26	No	No	Multi-technology	Wind, Small Hydro	Quantity	37	2	32
								29	17	1384
LEN	11° LEN	2010-12-17	No	No	Specific	Hydro, Biomass	Quantity	29	5	107
								49	2	2174
LEN	12° LEN	2011-08-17	No	No (*)	Multi-technology	Wind, Thermal (any)	Availability	31	18	2169
								31	1	450
LER	04° LER	2011-08-18	No	No	Multi-technology	Large Hydro, Small Hydro	Quantity	35	15	961
LEN	13° LEN	2011-12-20	No	No	Multi-technology	Wind, Biomass	Quantity	49	1	135
								49	10	823
LEN	15° LEN	2012-12-14	No	No	Multi-technology	Wind, Thermal (any)	Availability	49	3	252
								49	2	258
LER	05° LER	2013-08-23	Yes	No	Specific	Large Hydro, Small Hydro	Quantity	49	2	258
LEN	16° LEN	2013-08-29	No	No	Multi-technology	Wind	Quantity	25	8	1044
								53	6	497
LEN						Thermal (any)	Availability	53	10	633
								53	10	633

(continued on next page)

Table A.1 (continued)

Auction Type	Official name	Auction date	Auction with preliminary transmission capacity stage	Transmission risks allocated to generators	Auction products				Awarded projects	
					Technological banding	Technologies	Contract Type	Lead Time (months)	Number of projects	Capacity (MW)
LEN	17° LEN	2013-11-18	No	Yes	Multi-technology	Wind, Solar, Thermal (any)	Availability	26	3	377
LEN	18° LEN	2013-12-13	No	Yes	Multi-technology	Wind, Solar, Thermal (any)	Availability	53	20	1775
						Small Hydro, Large Hydro	Quantity	53	4	142
						Wind, Thermal (any)	Quantity	53	15	1049
LEN	19° LEN	2014-06-06	No	Yes	Multi-technology	Wind, Solar, Thermal (any)	Availability	31	3	398
						Large Hydro, Small Hydro	Quantity	31	1	418
LER	06° LER	2014-10-31	No	Yes	Specific	Wind	Quantity	36	3	370
						Solar	Quantity	36	11	827
LEN	20° LEN	2014-11-28	No	Yes	Multi-technology	Wind, Solar, Thermal (any)	Availability	50	8	764
						Large Hydro, Small Hydro	Availability	50	7	3865
						Wind	Quantity	50	3	44
LFA	03° LFA	2015-04-27	Yes	Yes	Specific	Wind	Availability	27	1	90
LEN	21° LEN	2015-04-30	No	Yes	Multi-technology	Thermal (any)	Availability	57	3	1607
						Large Hydro, Small Hydro	Quantity	57	10	355
LEN	22° LEN	2015-08-21	Yes	Yes	Specific	Wind	Availability	29	3	325
					Multi-technology	Thermal (any)	Availability	29	2	36
						Large Hydro, Small Hydro	Quantity	29	7	66
LER	07° LER	2015-08-28	No	Yes	Specific	Solar	Quantity	23	12	626
LER	08° LER	2015-11-13	Yes	Yes	Specific	Wind	Quantity	36	5	524
						Solar	Quantity	36	16	853
LEN	23° LEN	2016-04-29	No	Yes	Multi-technology	Thermal (any)	Availability	57	1	6
						Thermal (any)	Availability	57	6	188
						Small Hydro, Large Hydro	Quantity	57	14	194
LER	10° LER	2016-09-23	No	Yes	Specific	Small Hydro	Quantity	42	31	184
LEN	25° LEN	2017-12-18	Yes	Yes	Specific	Wind	Availability	37	1	69
						Solar	Availability	37	6	639
					Multi-technology	Large Hydro, Small Hydro	Quantity	37	2	12

Source: Own depiction based on ANEEL's auction notices, PPA drafts and CCEE (2021a).

(*) In this auction round, in case of delays in the transmission system, generators would receive contract revenues as normal, but they would need to procure compensatory firm energy certificates in order to honor the contract.

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